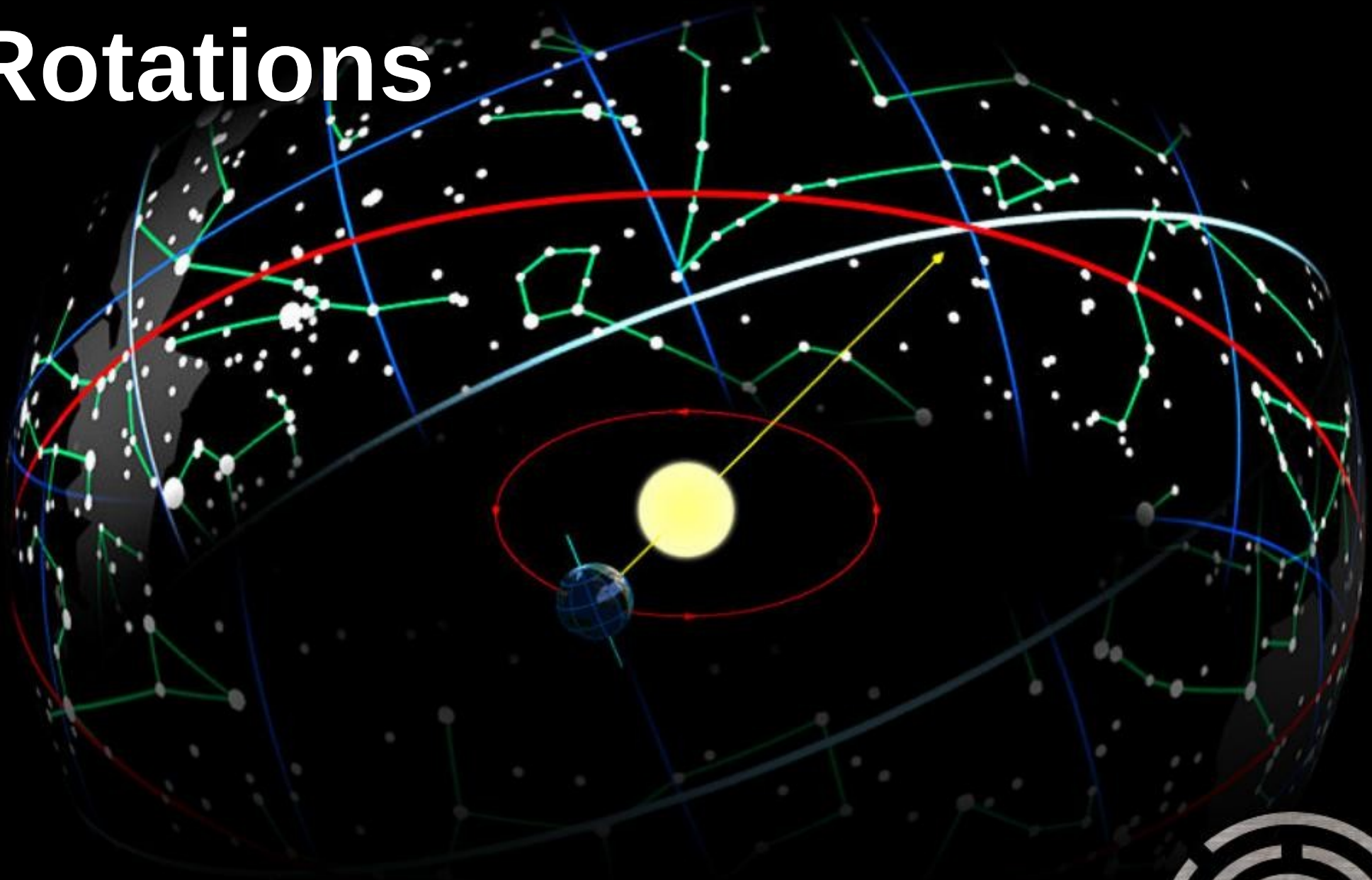
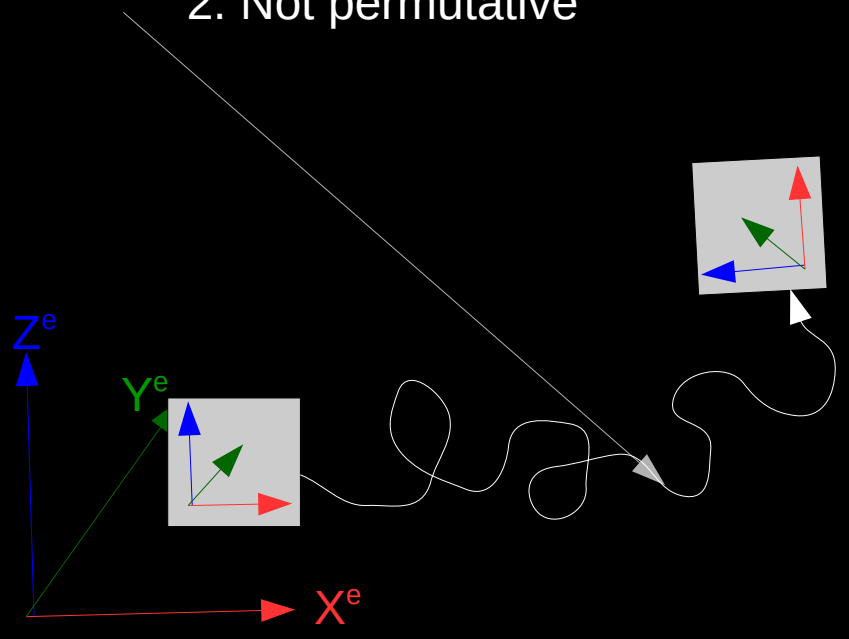
Rotations



Chain of Transformations

Chain Nx Δ Pose
Translation and Rotation
6DoF
Beware:
1. Δ Pose != Speed!
2. Not permutative



A chain (**not permutative**) of operations...

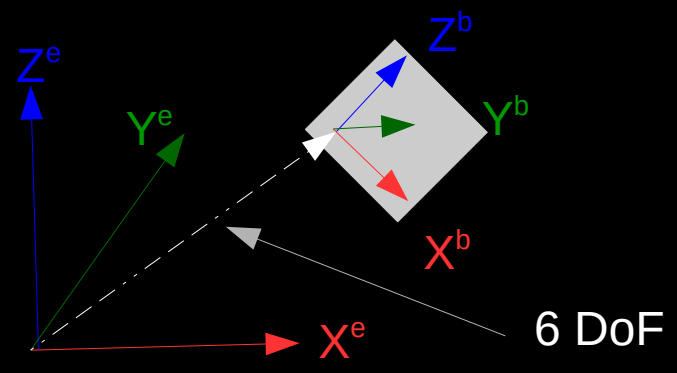
1) **Relative Nav**: Inertiale navigation, Koppelnavigation (dead reckoning):
I was in Pose-A and went through this chain of translations and rotations,
where am I now?

2) **Control**:
How do I get from Pose A to Pose B?

3) **Predictor**:
at this 6**D**-speed,
where will I be in T seconds?
f(Pose, d Pose/dt) ,
what will be my Δ Pose

4) **History**:
If I was in Pose-A war and am now in Pose-B: How did I move?

Translation



Translation is not a big science

A → B

Translation vector B-A = $(X_b - X_a \quad Y_b - Y_a \quad Z_b - Z_a)$ = $\mathbf{T^a_b}$

A → B → C → D:

I get to A + T^a_b + T^b_c + T^c_d

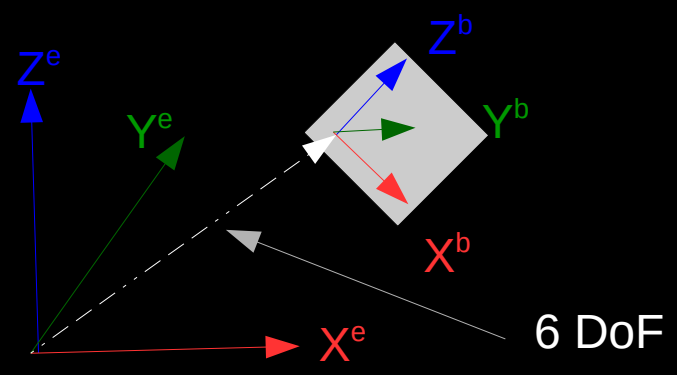
Sequence can be arbitrary! Yes, translation is **permutative**

A, B, C, D, T^x_y : 3D Vectors

Distance A-B:

Distance = sqrt($(X_b - X_a)^2 \quad (Y_b - Y_a)^2 \quad (Z_b - Z_a)^2$)

Translation : simply add 3D vectors, and is permutative



Rotations

Simply add?
(0, 0, 90) +
(0, 90, 0) +
(90, 0, 0) +
(0, 90, 0) =

(90, 180, 90) ? **NO!!!**
Demo sequence x, y, z / z, y, x

Rotations …. on the other hand ….
simply add? only in 1D
but not in 3D

Rotations, very tricky

Paper Airplane: Permutations (+90 Pitch, +90 Roll), at the same time

Relative to airplane

Relativ to earth (own body axes)

Ambiguous: Pitch 90° : What is Roll and what is Yaw?

Roll 90° : What is Pitch and what is Yaw?

(90°Z + 90°X) - (90°Z + 90°X) != not back!

Example with paper plane

Attitude = Integration/Sum of 3D Gyros measurements?

the gyros also rotate!

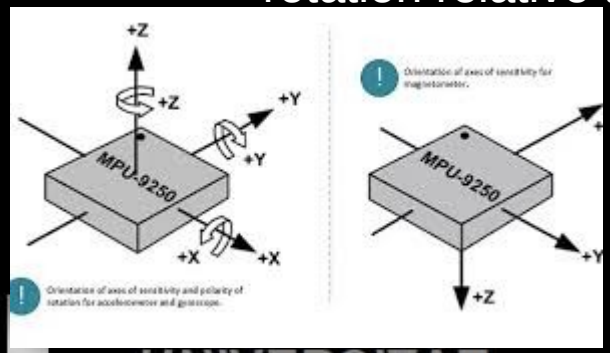
e. g. Z axis is no longer vertical

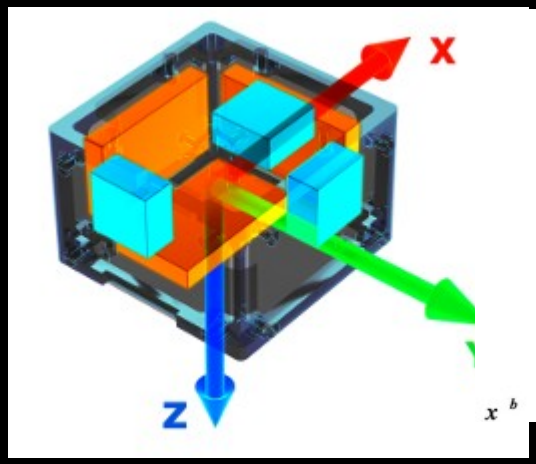
Gyro says : (2°, 1°, 3°)/s (*t) how do I „add“ this?!?

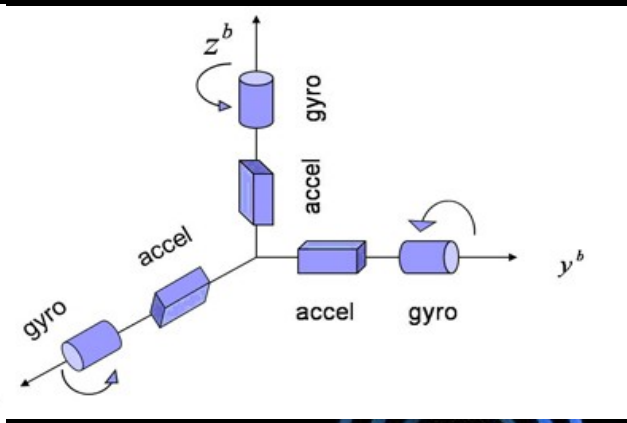
3 Axes IMU

Fixed, turns together with plane!

-> rotation relative to body frame







Not permutative!

But I have 3 measurements: Pitch, Roll, Yaw – and must sich integrate sequentially (not add!)

With every rotation in 1 axis the other ones change!

If I ignore that, I accumulate an enormous amount of error.

which one first?... No matter how, it is wrong!

but with super small steps and small angles, the error is minimum

für jede Achse: zum berechnen Duchnitt von anfang und end-Lage

aviation standard:

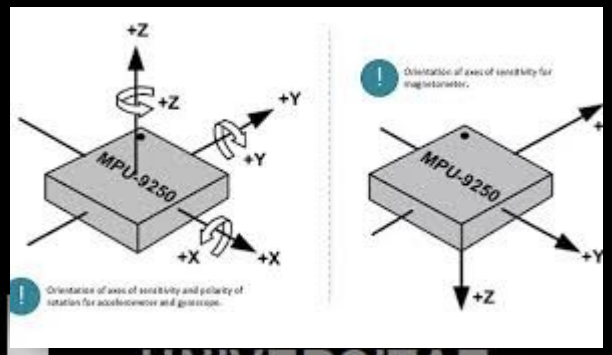
RPY: Roll-Pitch-Yaw

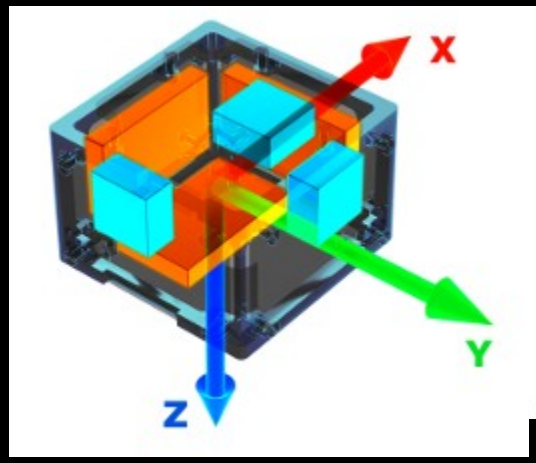
first Yaw, than Pitch and finally Roll

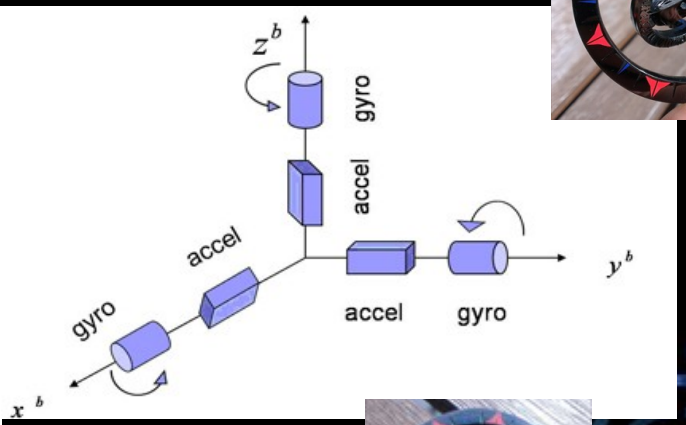
(Why RPY and not YPR? With multiplication the first is right)

Gimbal construction corresponds.

exterior Roll (X), than Pitch (Y) than Yaw (Z)







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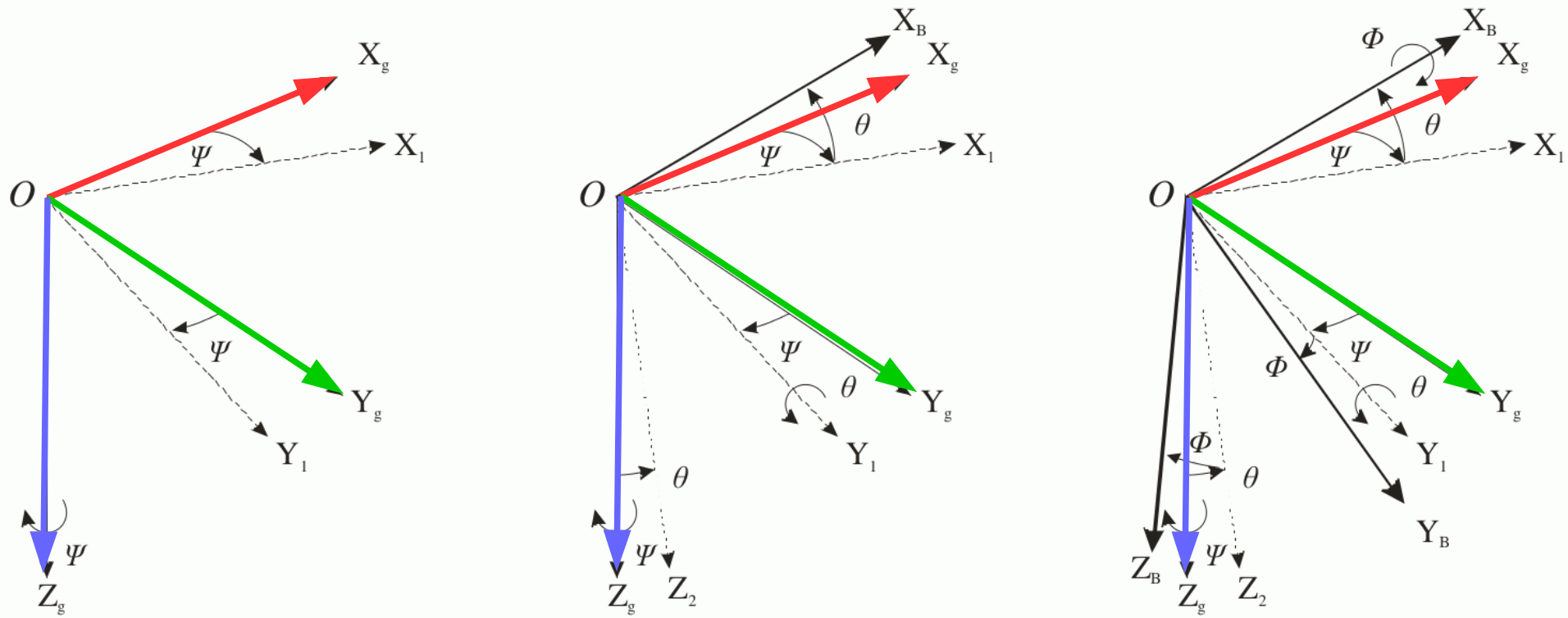
Rotation in one axis chages the other axes

RGB
XYZ -> RPY

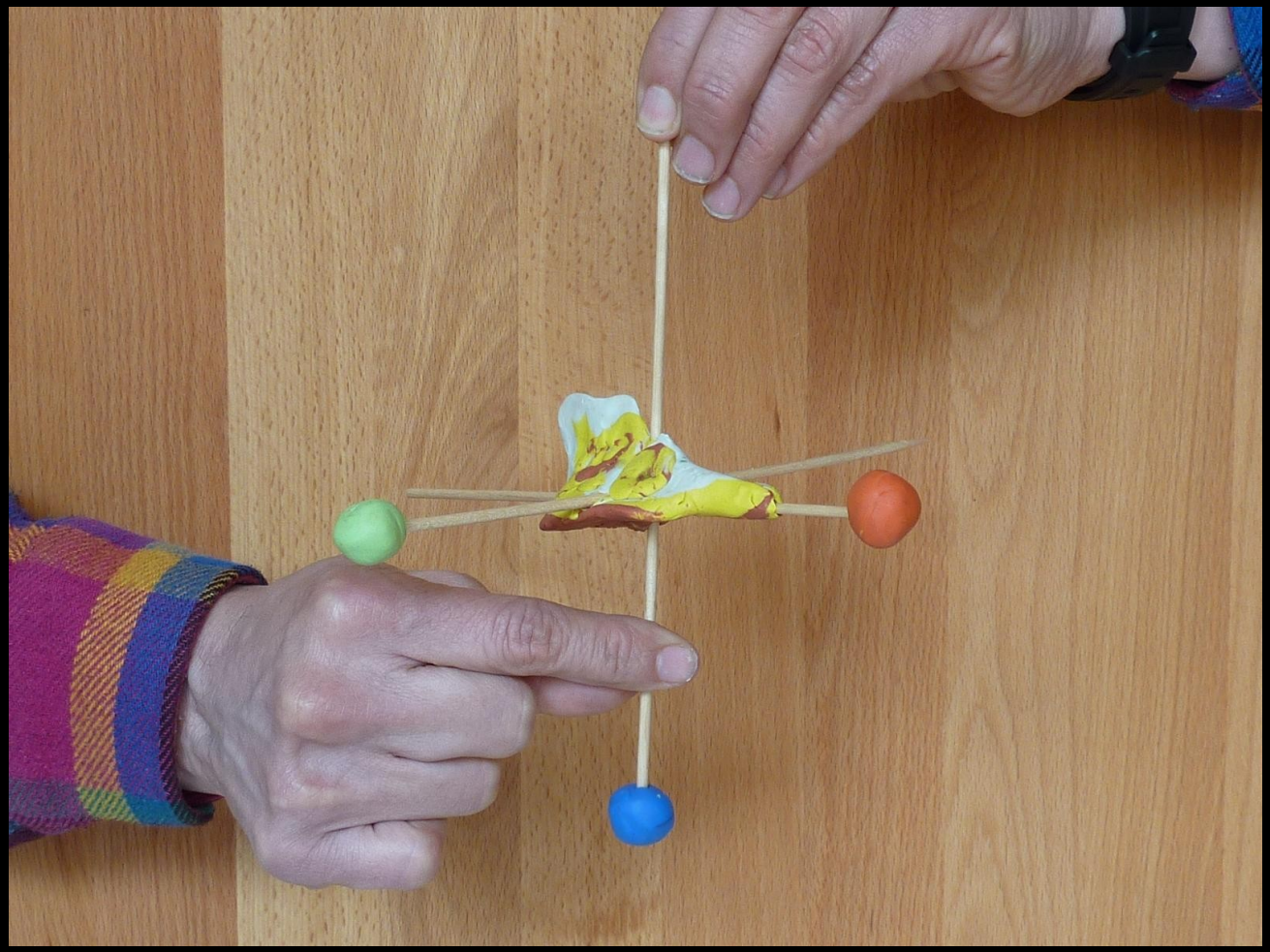
X Rot → Roll
Y Grün → Pitch
Z Blau → Yaw



first Yaw (Z) than Pitch (Y) than Roll (x)

Finally every axis moved around two angles
(3 combinations of 2 out of 3 angles)

Example of Rotation

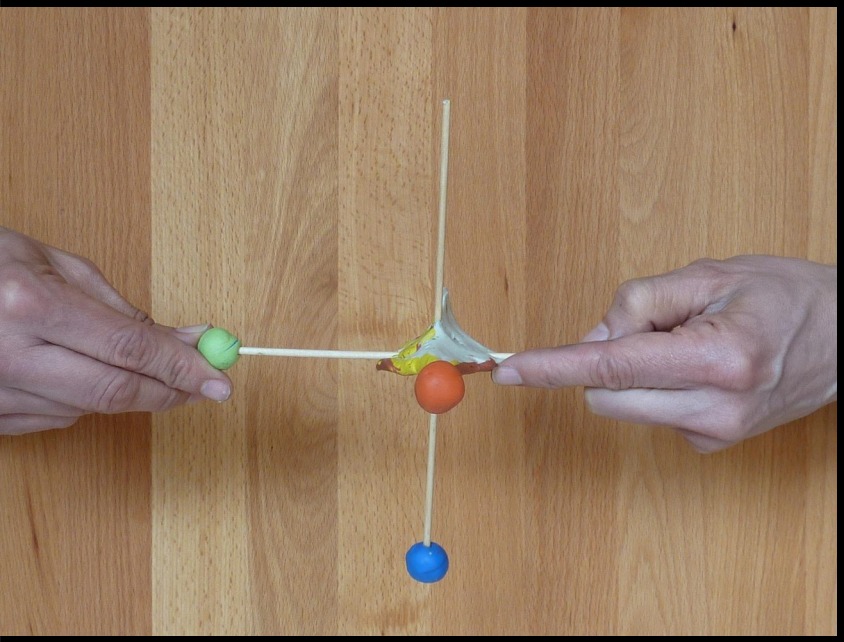


RGB
XYZ -> RPY

X Red → Roll
Y Green → Pitch
Z Blue → Yaw

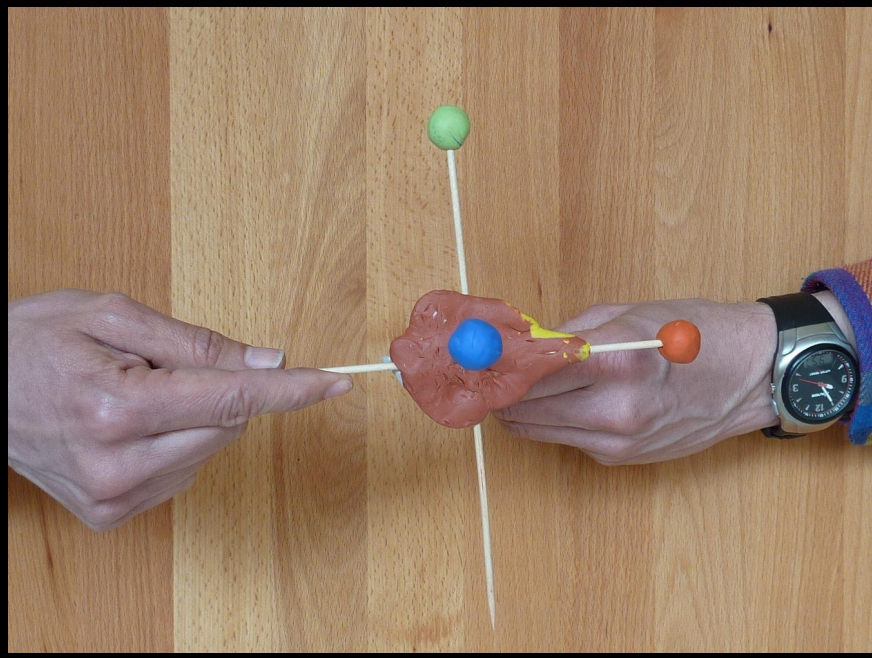
Beware! Standard order
RPY
1. Yaw
then 2. Pitch
then 3. Roll
(from right to left for multiplication (concatenation))

Example of Rotation
Pitch up, Yaw left



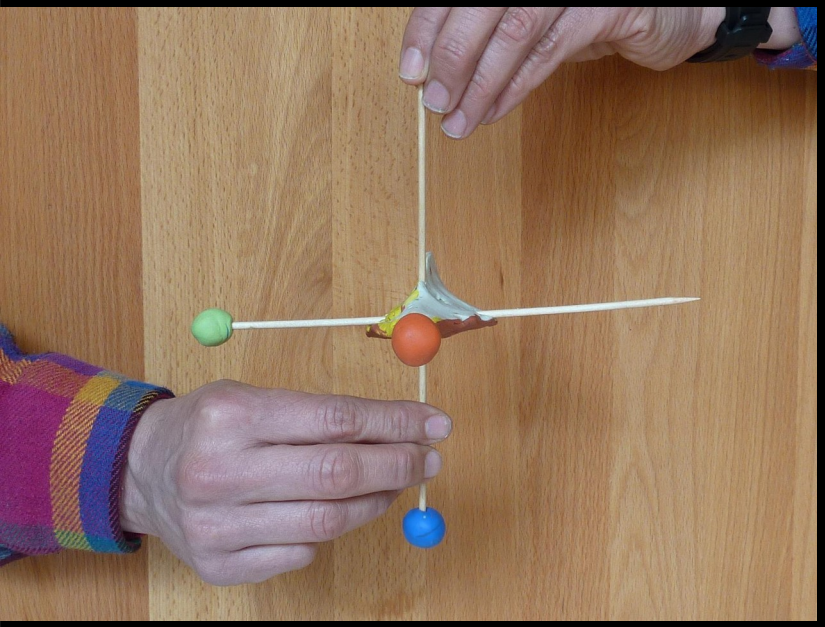
Pitch Yaw

90^0 (0,1,0)
90^0 (0,0,1)



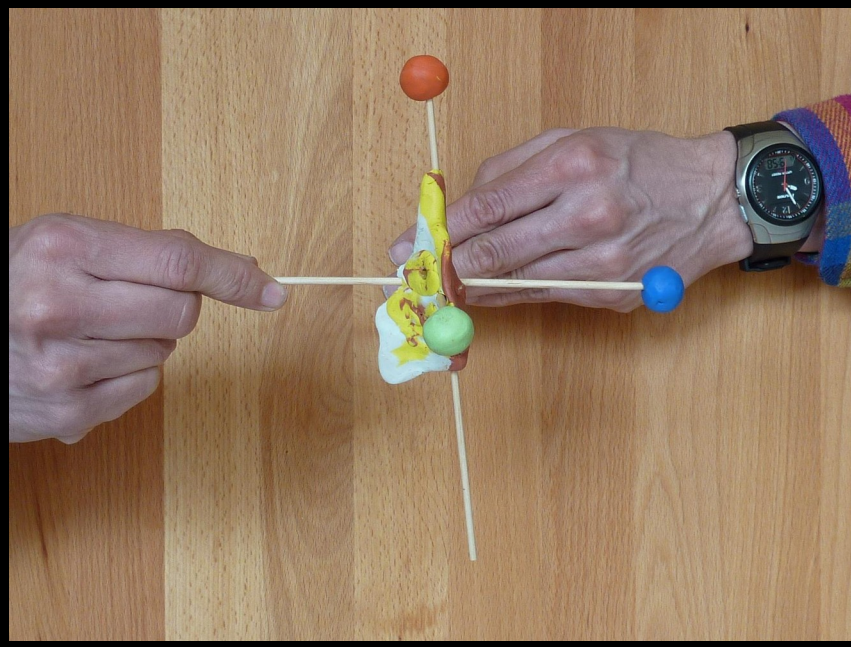
= Rotor (0,0,1) * Rotor (0,1,0)
(Multiplication from right to left)

Example of Rotation
Yaw left, Nick up



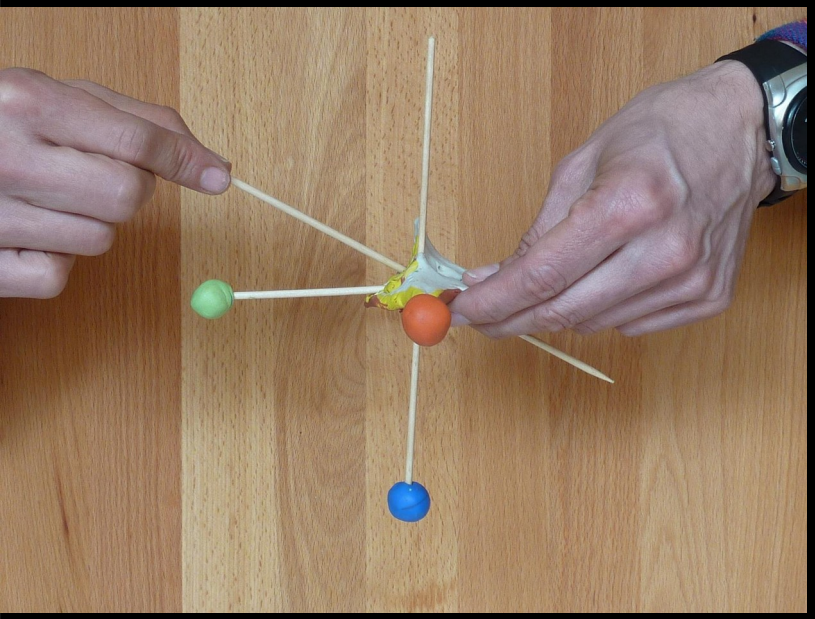
Yaw Pitch

90^0 um (0,0,1)
90^0 um (0,1,0)



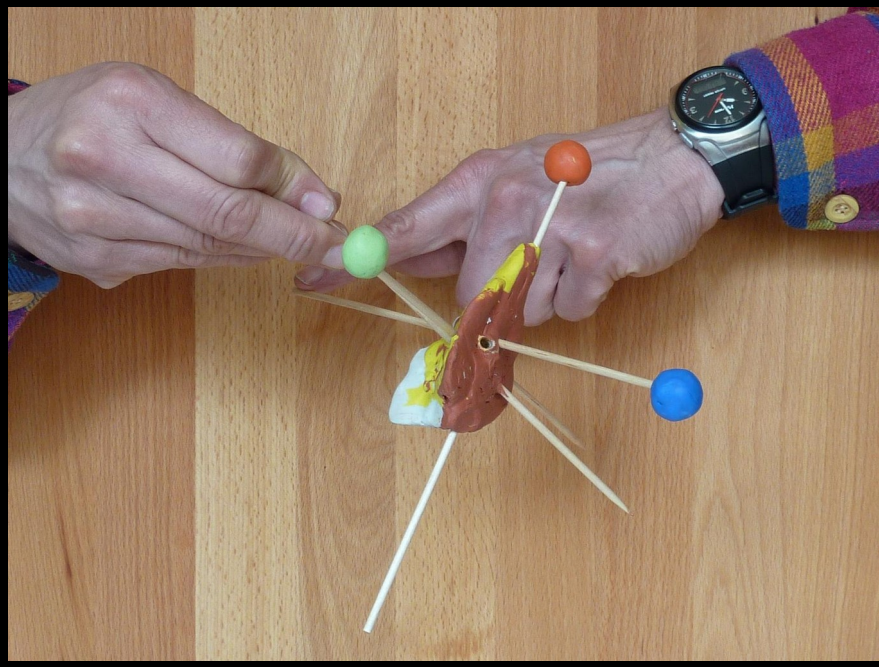
= Rotor (0,1,0) * Rotor (0,0,1)
(Multiplication from right to left)

Example of Rotation
Yaw and Nick together

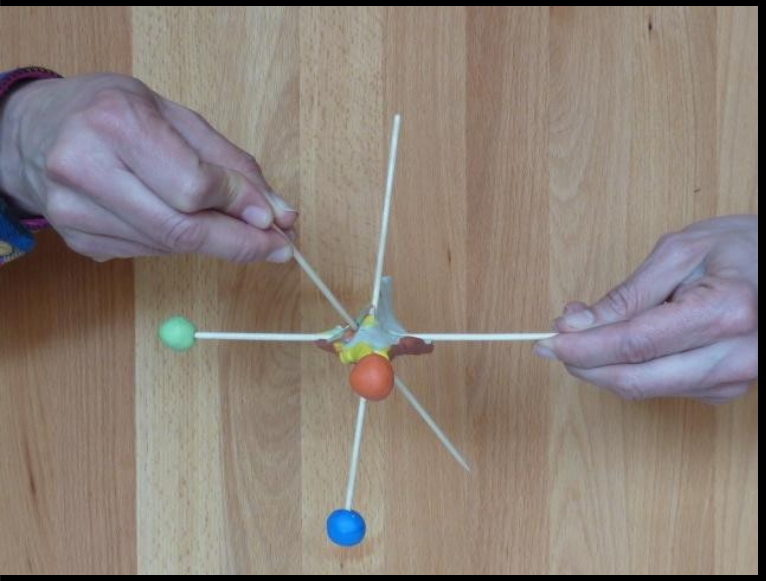


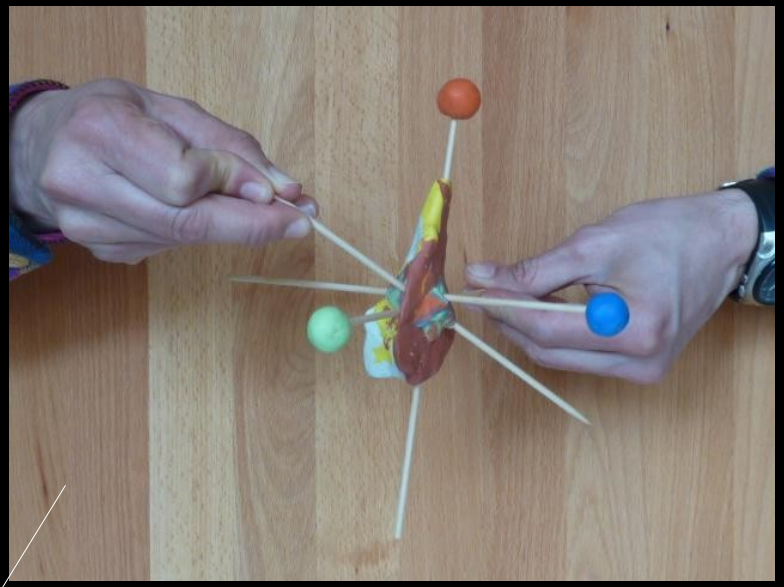
Both at the same time (0,1,1) Axis

But how many degrees?

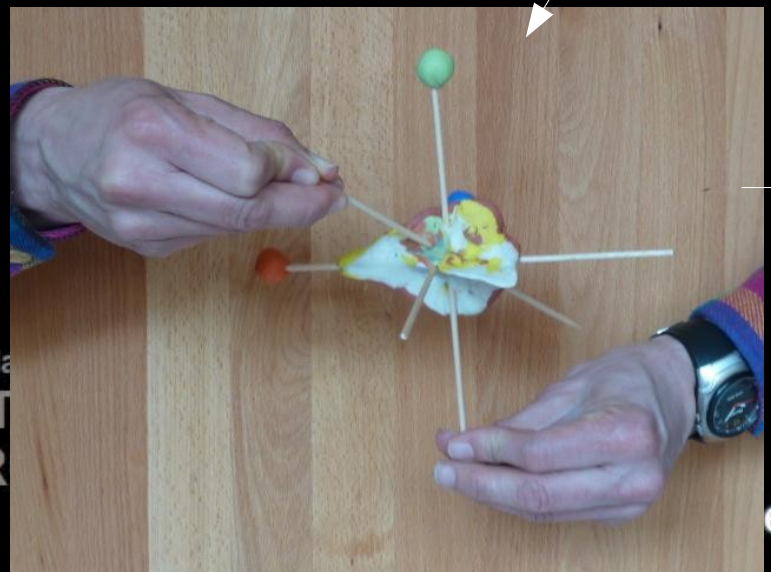


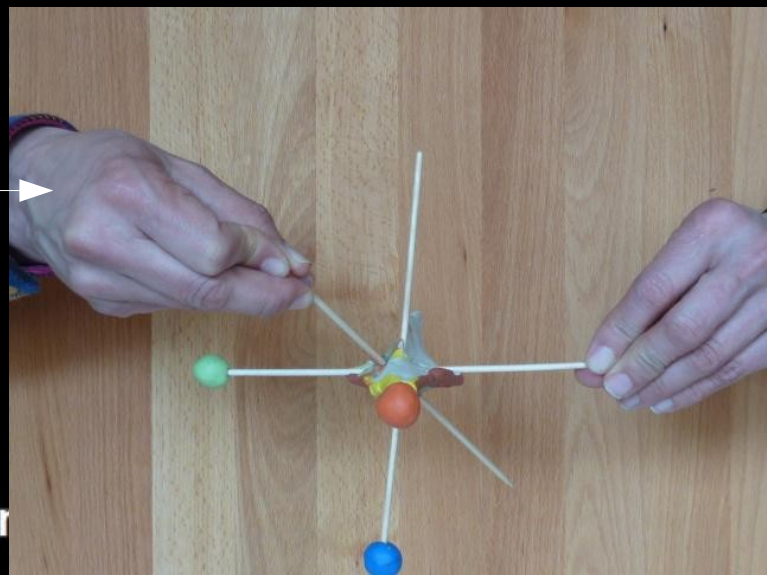
Example of Rotation
Yaw and Nick and Roll together





soryr (1,1,-1)





3x (1,1,1) axis
by 120°

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e Inforr

Rotation around the other axis
until we permute axes

(1,0,0) : 4x 90°: we are at the beginning
(1,1,1) : 3x 120° == 4x 90°
(1,1,0) : 2x 180°

Pitch: Y Roll : X

90^0 (0,1,0) → 90^0 (1,0,0) = 120^0 (1 1 1) !!

120^0 (1 1 1) = Zyclic permuation of the 3 axes

Siehe auch Würfel (sat)

Siehe Filme
player-rotationen
Pause Kardanischeaufh.

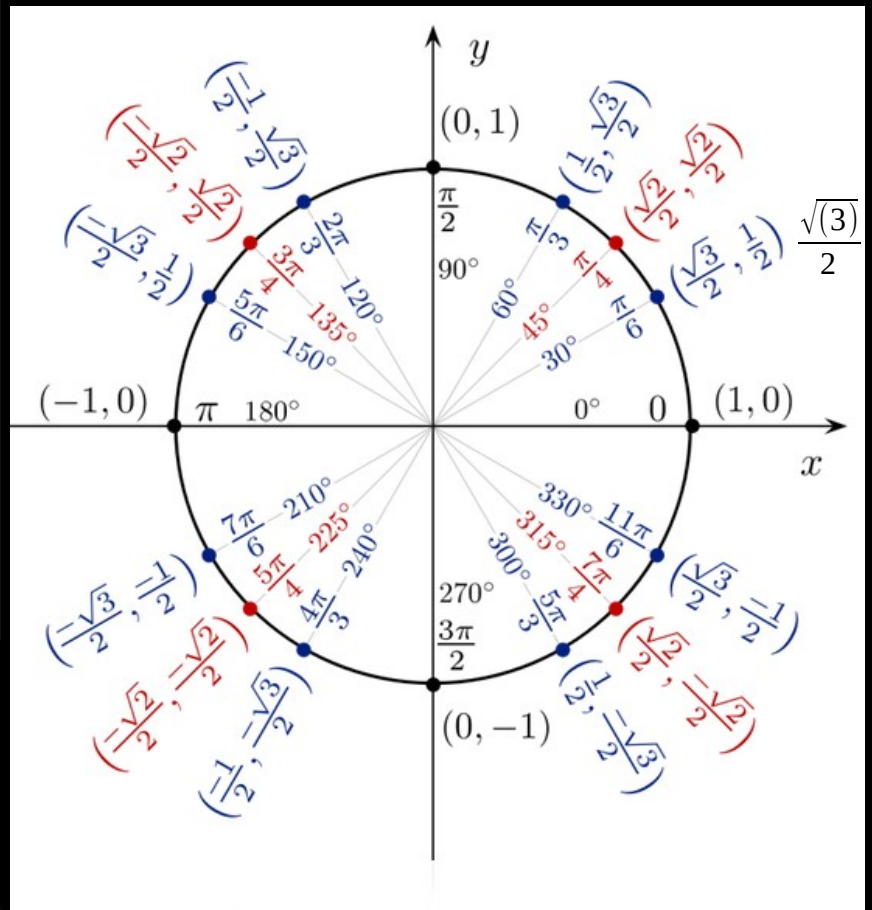
Representation of angles

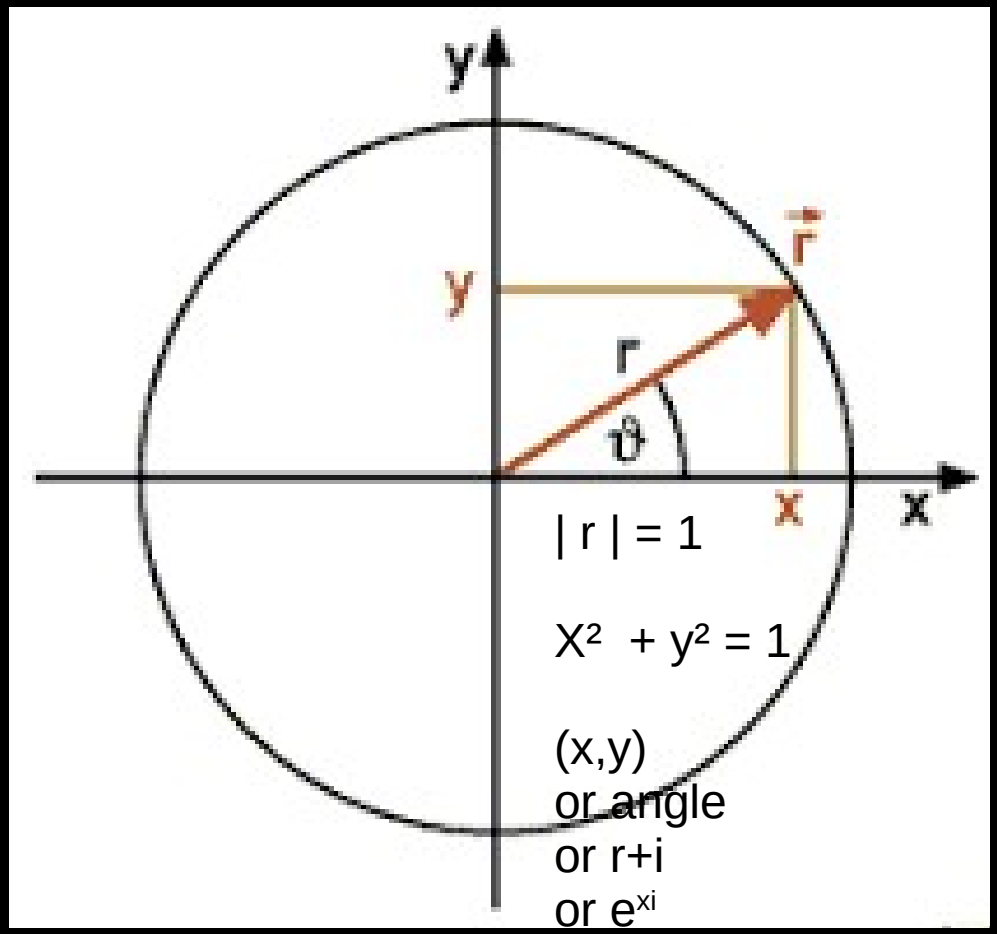
e.g. in 2D (flat)

angle	**(polar)**
(x,y)	**(Cartesian)**
r+i	**(Cartesian in complex plane)**
e^{xi}	**(complex plane, add movement: e^{txi}**

(e.g. 2D) Why real + “i”? Why not 2 Coordinates?

“i” instead of real y?
But why?
And then | 2 + 2i | = ??





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Why real + “i”?
Why not 2 Coordinates?

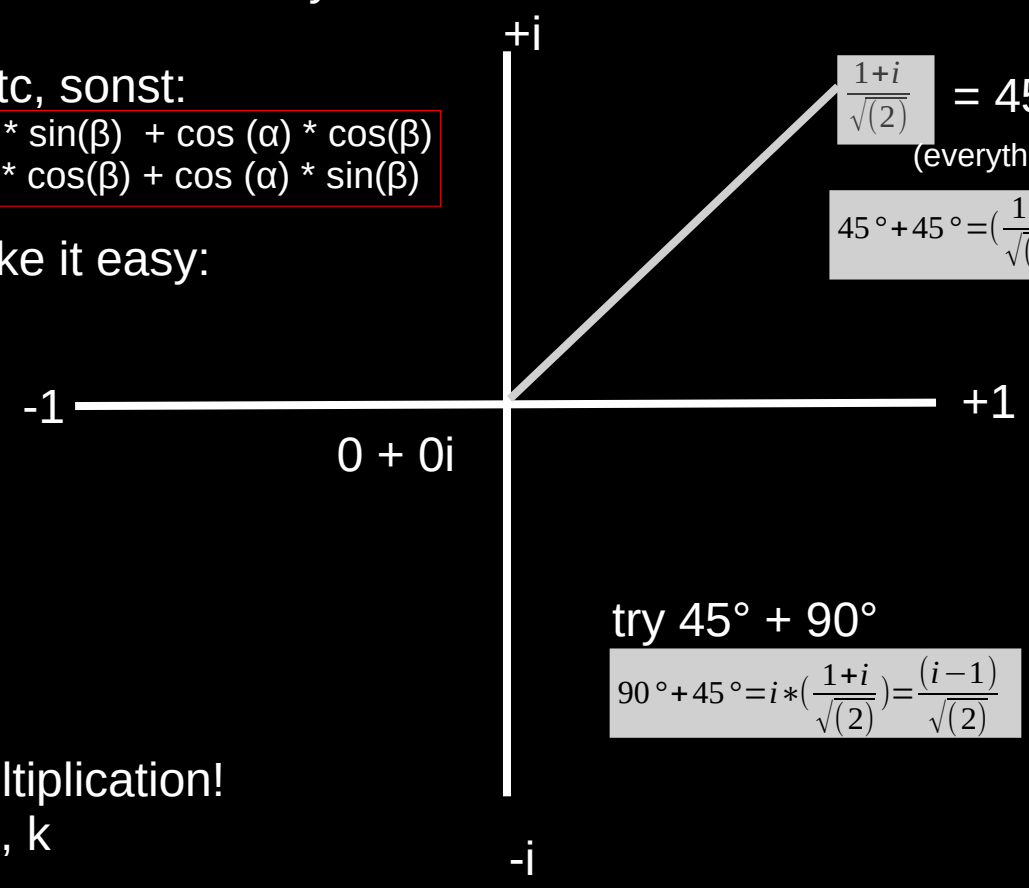
Rotations in Cartesian coordinates by multiplications!
without sin, cos, etc etc, sonst:

$\cos(\alpha + \beta) = -\sin(\alpha) * \sin(\beta) + \cos(\alpha) * \cos(\beta)$
$\sin(\alpha + \beta) = \sin(\alpha) * \cos(\beta) + \cos(\alpha) * \sin(\beta)$

complex numbers make it easy:

w2 * w1

W1 & w2 are complex numbers



$\frac{1+i}{\sqrt{(2)}}$ = 45°

(everything normalized)

$$45° + 45° = \left(\frac{1+i}{\sqrt{(2)}}\right) * \left(\frac{1+i}{\sqrt{(2)}}\right) = i = 90°$$

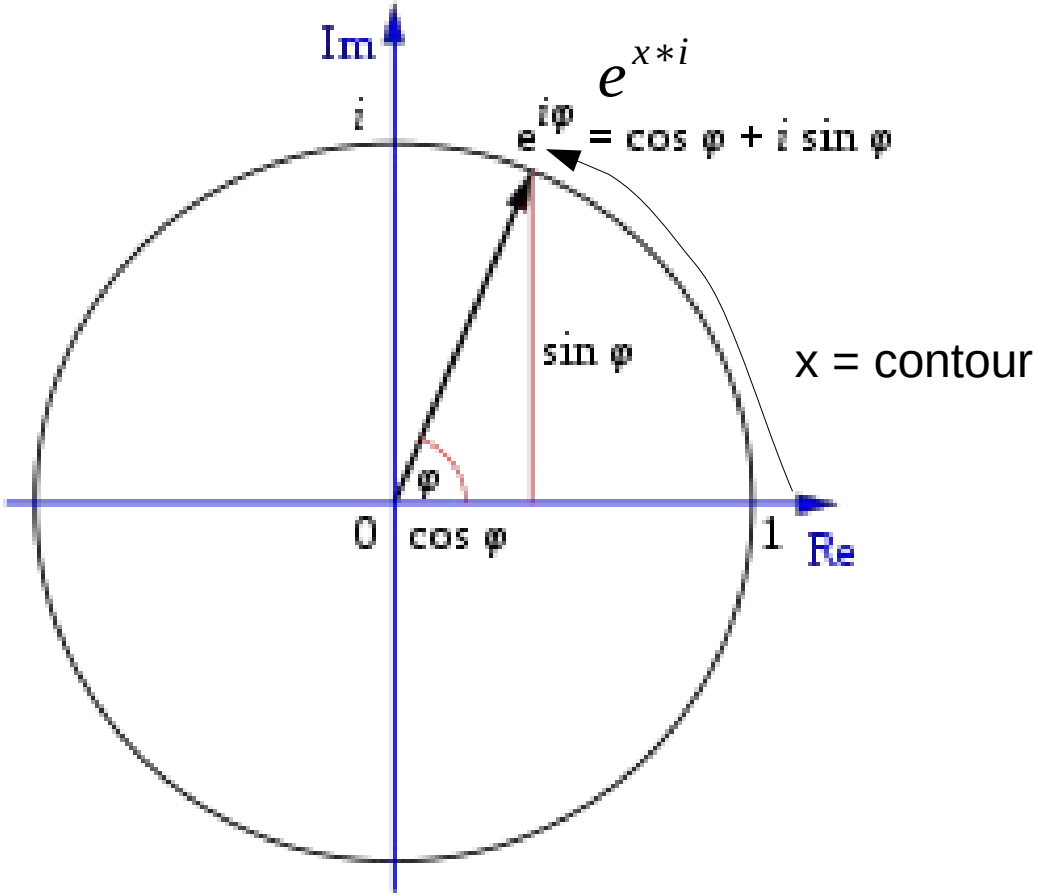
1 + i *
1 + i
_____
-1 + 2i + 1
= 2i

try 45° + 90°

$$90° + 45° = i * \left(\frac{1+i}{\sqrt{(2)}}\right) = \frac{(i-1)}{\sqrt{(2)}}$$

-1 * 1 = 180°
i * 1 = 90°
-> Rotate by multiplication!
now in 3D: 1, i, j, k

$e^{(xi)}$: Euler's Formula



x = contour

Nice for movements
eg. rotating objects

angle = e^{txi}

./player-e-hoch-ti

Remember: Alternatives

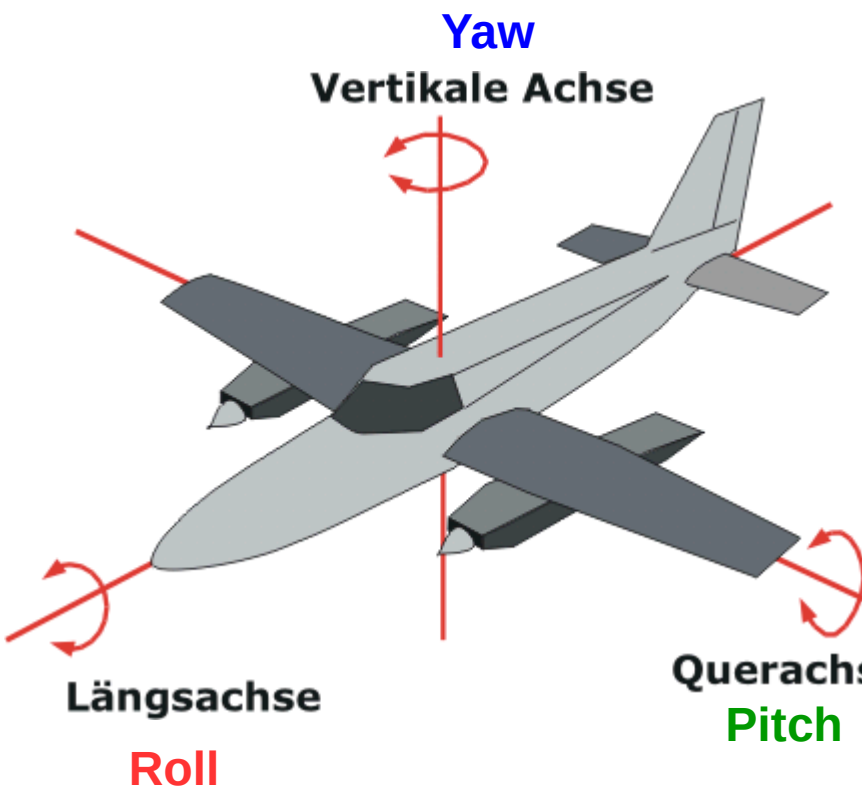
How to start?

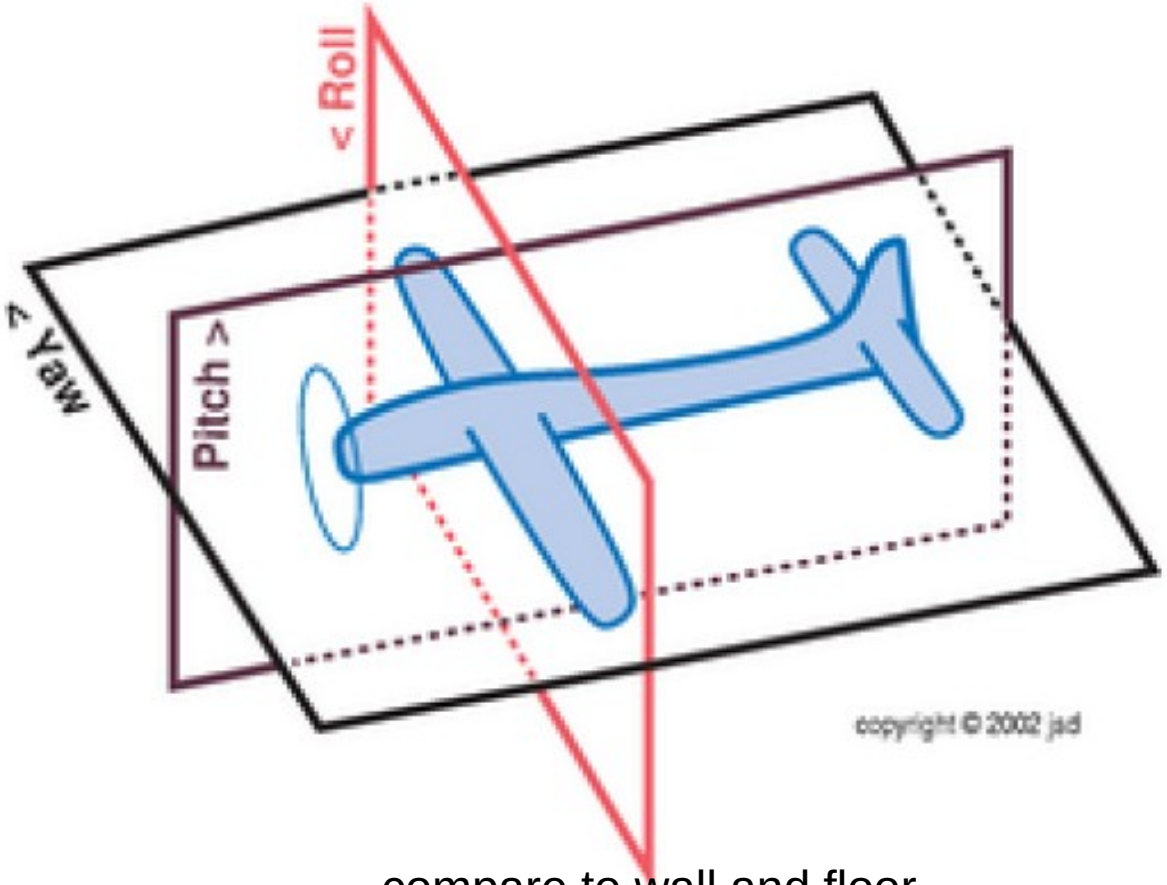
angle	(polar)
(x,y)	(Cartesian)
r+i	(Cartesian in complex space)
e^{xi}	(complex plane)

coordinate system
- Cartesian
- Cylindrical
- Polar

Calculation:
- Sequence of RPY
- Rotation matrices around X, Y, Z
- around vector
- On plane

Rotation: around Axes or on Surface

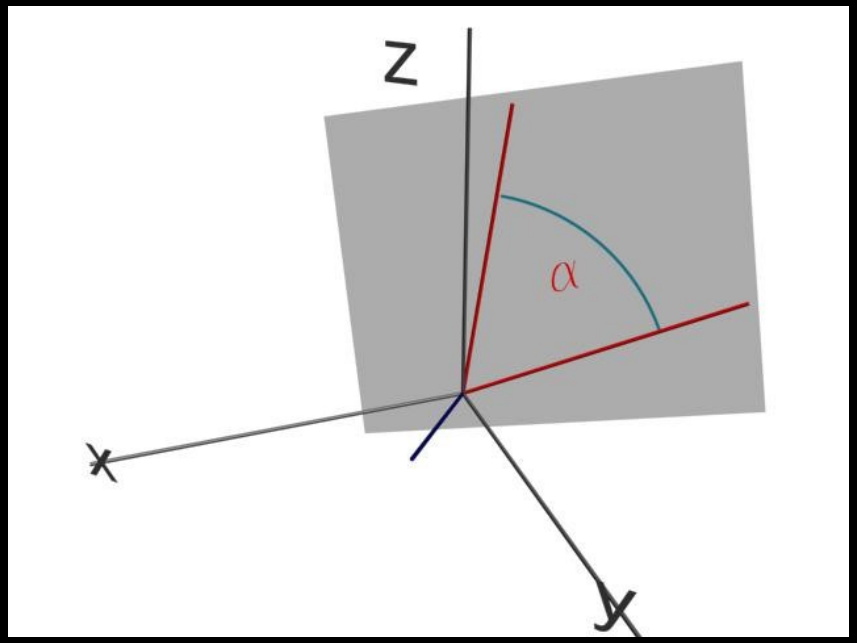




compare to wall and floor

Surface is defined by two non-collinear vectors
But... Many vectors may define the same surface
(or one (rotation vector) orthogonal to surface)

Rotation around any Axis



Surface:

2 Vectors define the surface
angle defines rotation angle

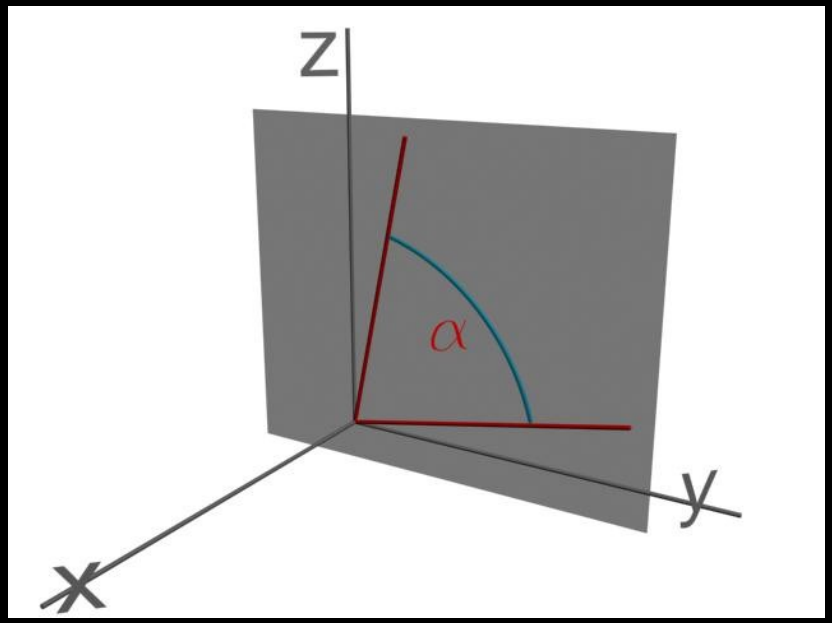
Be careful with collinear vectors!

Rotation by 0° and 180°
(Is it a problem?)

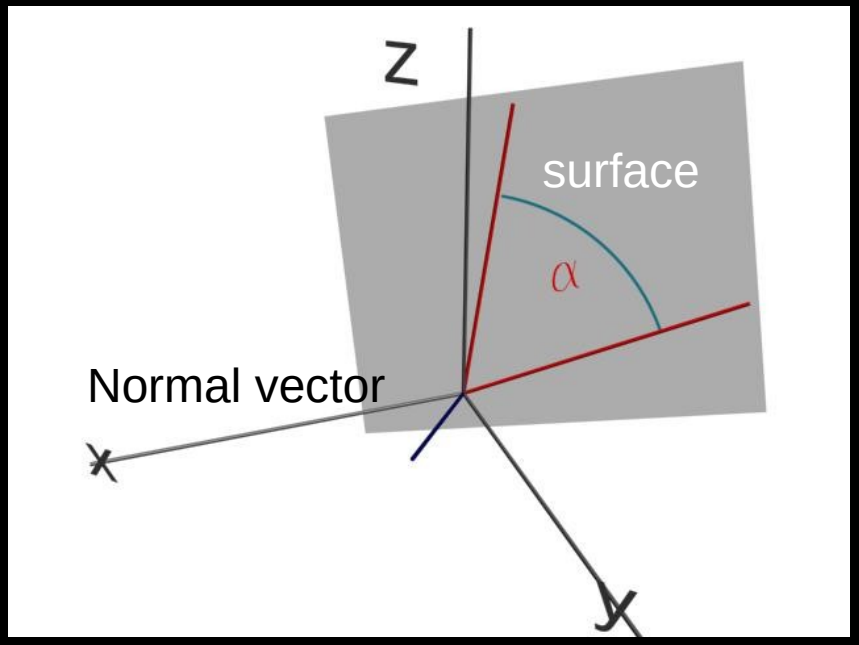
Rotation vector:

Vector defines the axis
Rotation angle:
extra parameter
or
length of vector (0 .. 1)

Rotation: Axes or Surfaces



surface parallel to Z-Y
Axis: X



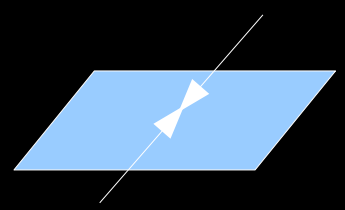
Rotation axis is the most effectiv
I don‘t go first in X and then in Y
I walk the direct vector

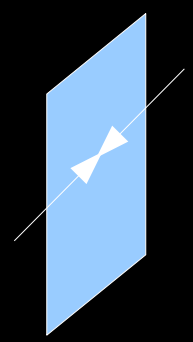
Mein Holz Beispiel

Rotations: ½ Angles?
Problem: Rotate 180°

180°?... we simply fly back: from (1,0,0) to (-1, 0, 0)
from v to w... but v & w are collinear
v X w = 0... Rotation vektor (0,0,0)! might be anything!

let's try by 180°, roll, pitch, yaw : all the same?





Vectors are collinear !

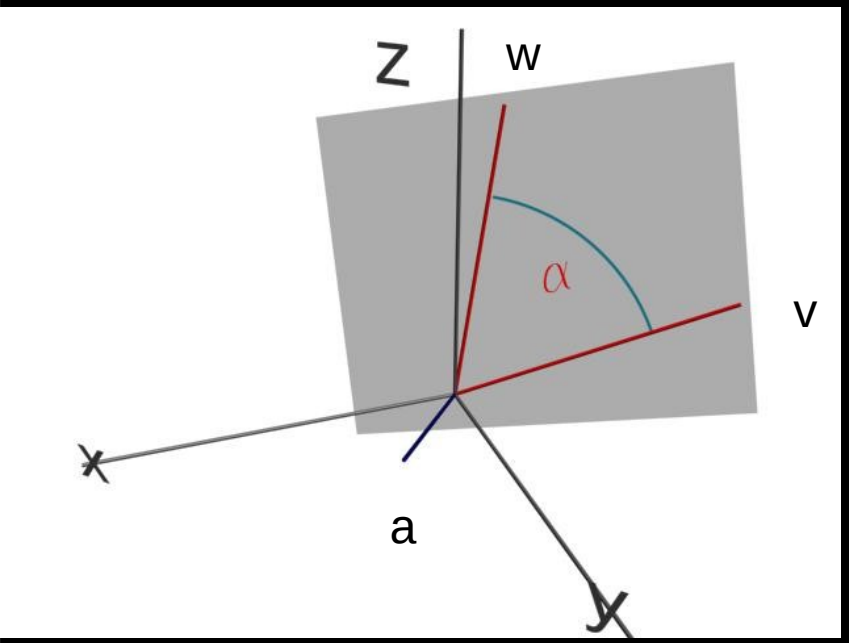
Zeigen mit zwei Stiffte
3 Achsen + Papierflieger

Rotational Axis & Plane

Axis:
orthogonal to plane v & w

$\vec{a} = \vec{w} \times \vec{v}$ from v to w

$-\vec{a} = \vec{v} \times \vec{w}$ from w to v

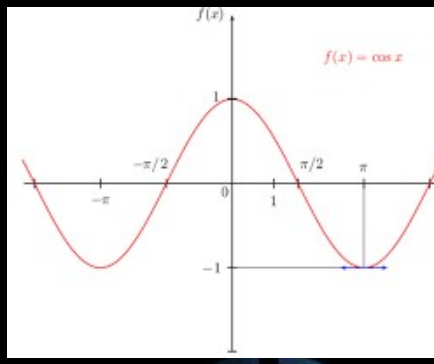


$\cos \alpha = \vec{v} \bullet \vec{w}$

v & w normalized

v^2 and $w^2 = 1$

$v \bullet w = w \bullet v$
$\cos(\alpha) = \cos(-\alpha)$
Back an forth only
via axis/-axis



Ambiguous?
We only use positive angles
and the direction is set by vector a

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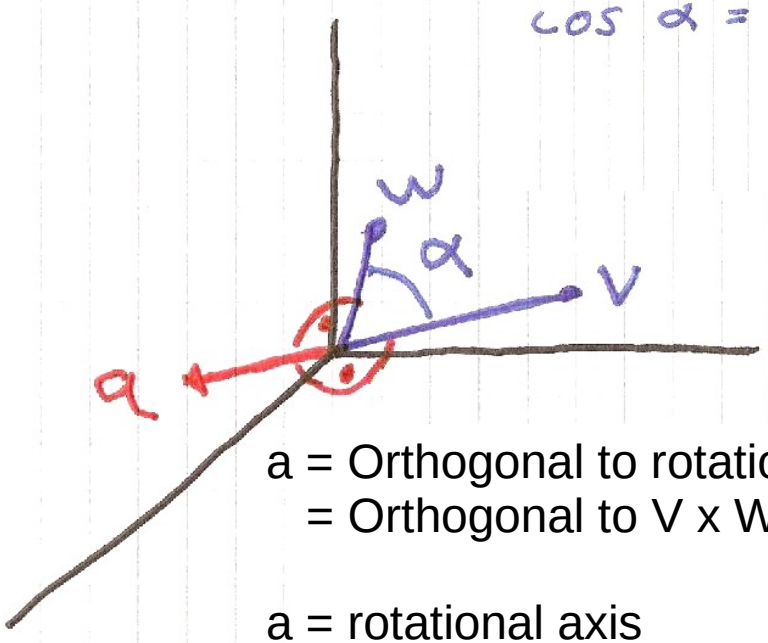
Rotational Axis

from W to v

$$\cos\alpha = \frac{v \cdot w}{|v| \cdot |w|} = \frac{v_1 w_1 + v_2 w_2 + v_3 w_3}{\sqrt{v_1^2 + v_2^2 + v_3^2} \cdot \sqrt{w_1^2 + w_2^2 + w_3^2}}$$



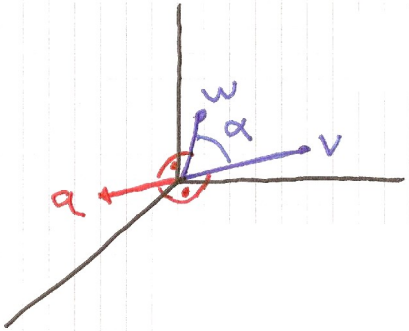
a = Orthogonal to rotation plane
= Orthogonal to V x W

a = rotational axis

$$a = v \times w = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$$

“a” Relative to what?

Rotational Axis



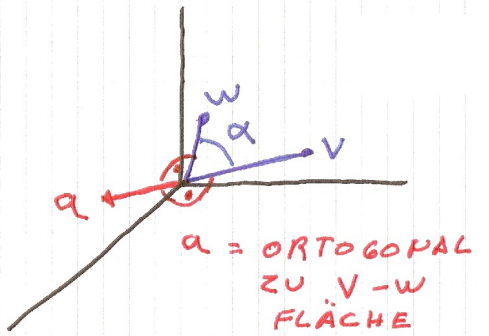
$$\cos\alpha = \frac{v \cdot w}{|v| \cdot |w|} = \frac{v_1 w_1 + v_2 w_2 + v_3 w_3}{\sqrt{v_1^2 + v_2^2 + v_3^2} \cdot \sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$a = v \times w = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}$$

$$= \begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} w_1 \\ w_2 \\ w_3 \\ w_1 \\ w_2 \\ w_3 \end{array} = \begin{pmatrix} v_2 \cdot w_3 - v_3 w_2 \\ v_3 \cdot w_1 - v_1 w_3 \\ v_1 \cdot w_2 - v_2 w_1 \end{pmatrix}$$

v X w = -(w X v)

Rotational Axis Example



a = ORTOGONAL ZU V-W FLÄCHE

a = DREH ACHSE

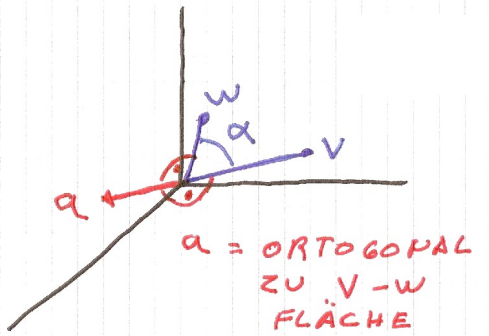
z.B. $\begin{pmatrix}1\\0\\0\end{pmatrix} \rightarrow \begin{pmatrix}0\\1\\0\end{pmatrix}$

$$\cos\alpha = \frac{0}{1\cdot 1} \Rightarrow \alpha = 90^\circ$$

$$a = \begin{array}{c} \begin{matrix} 1 & 0 \end{matrix} \\ \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \begin{matrix} 0 & 0 \end{matrix} \end{array} = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

is this correct?

Rotational Axis Example



z.B. $\begin{pmatrix}1\\0\\0\end{pmatrix} \rightarrow \begin{pmatrix}0\\1\\0\end{pmatrix}$

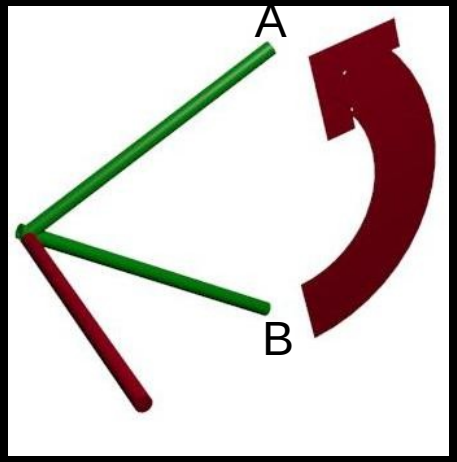
$$\cos\alpha = \frac{0}{1\cdot 1} \Rightarrow \alpha = 90^\circ$$

$$a = \begin{array}{c} \begin{matrix} 1 & & 0 \end{matrix} \\ \begin{bmatrix} 0 & & 1 \\ 0 & & 0 \\ 1 & & 0 \\ 0 & & 1 \end{bmatrix} \\ \begin{matrix} 0 & & 0 \end{matrix} \end{array} = \begin{pmatrix}0\\0\\1\end{pmatrix}$$

Siehe Beispiel Prog
rotations-achse-winkel.cpp

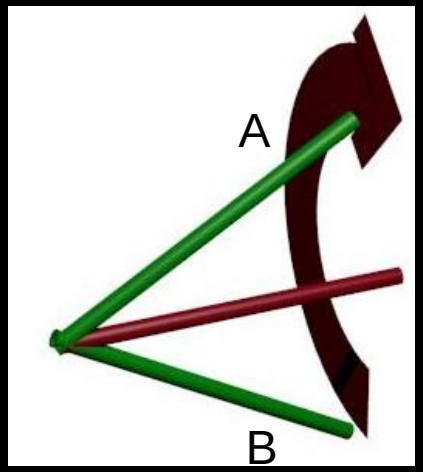
Warning: This was (010) to (100)
Try also (100) → (010)

But... There are other Axes!



Shortest way:
acos(B * A)

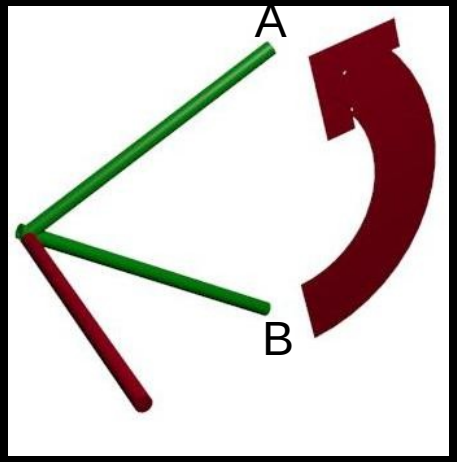
Axis = B X A



Longest way
180°

Axis = (A + B)/2

But… There are other Axes!



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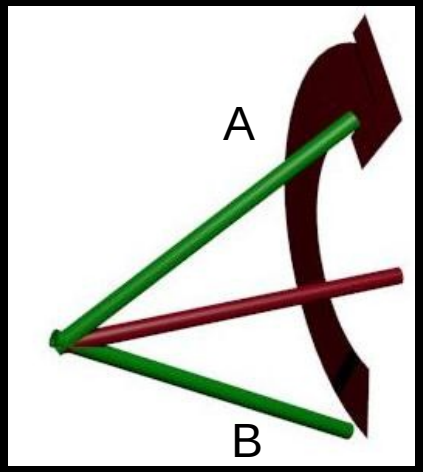
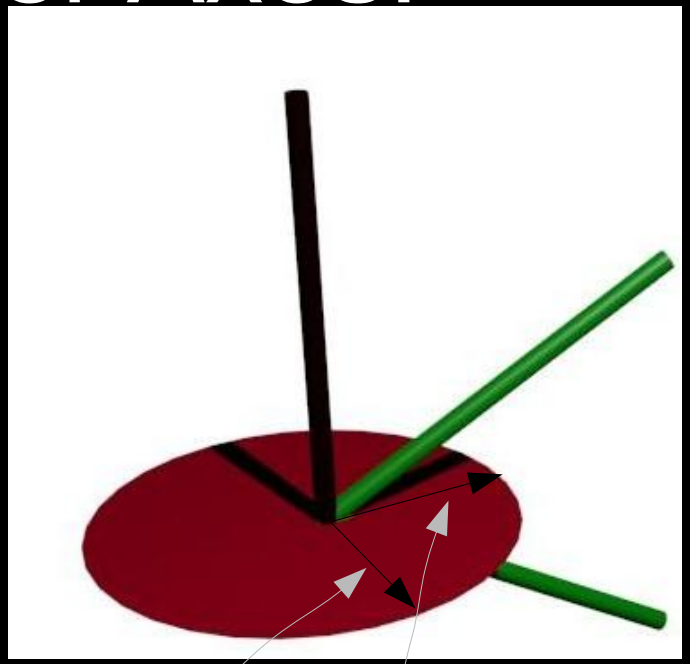


Longest way
180°

Axis = (A + B)/2

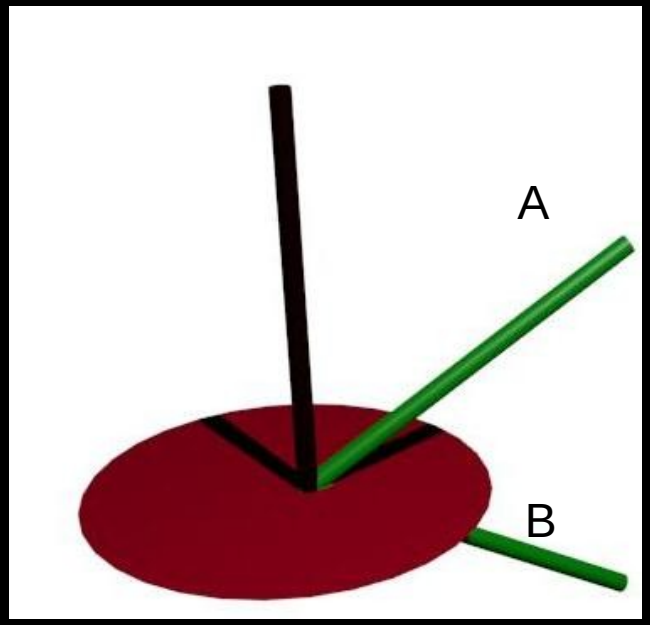


All vector on this plane
would also work.

Plane is defined by vectors

A X B
(A + B)/2
Axis = (A X B) X ((A +B)/2)
Or all vectors orthogonal to
A-B

But... There are other Axes!



All vector on this plane
would also work.
Orthogonal to
A - B

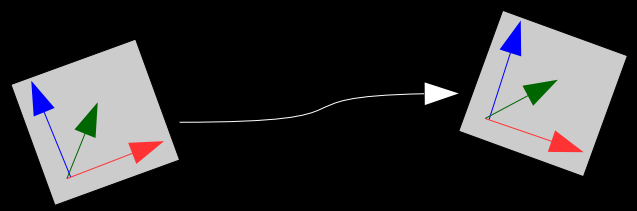
Which one shall I use?

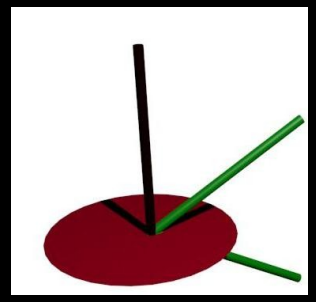
one degree of freedom:
it is all the same

no degree of freedom
there is only one

Unambiguous with 2 (or 3) Vectors

$(a_x \quad a_y \quad a_z) \quad \rightarrow \quad (b_x \quad b_y \quad b_z)$

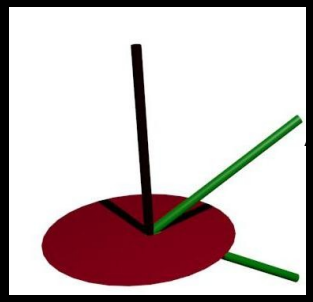




There is always exactly **one** rotational axis, which transfers the systems into each other.

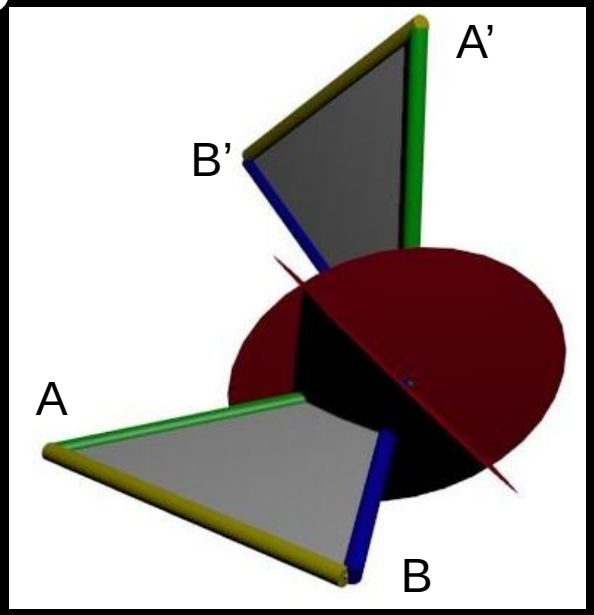
Or “Direction Cosinus Matrix”

Unambiguous with 2 vectors



one vector
one plate

All orthogonal to
A - A’



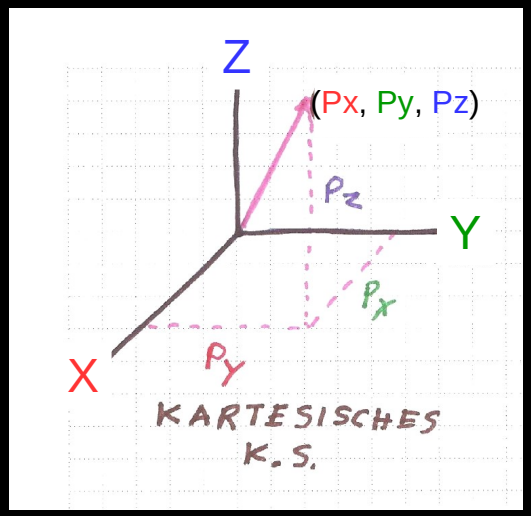
two vectors
two plates

cross section of both plates
-> 1 vector

one orthogonal to
A – A’
and to
B – B’

so:
(A-A’) X (B -B’)

Remember def. of coord. Sys



$$\vec{P} = (P_x \quad P_y \quad P_z)$$

Better like this:
(in order to multiply with matrices)

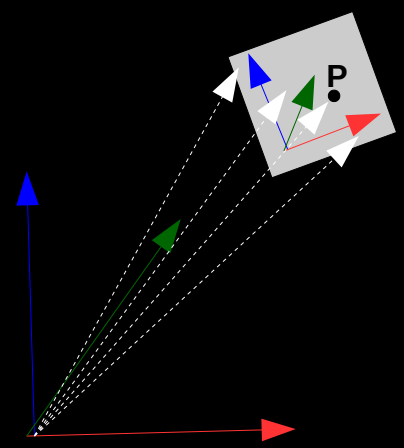
$$\vec{P} = \begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix}$$

$$\vec{P} = P_x * \vec{e}_x + P_y * \vec{e}_y + P_z * \vec{e}_z$$

$$\vec{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad \vec{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad \vec{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

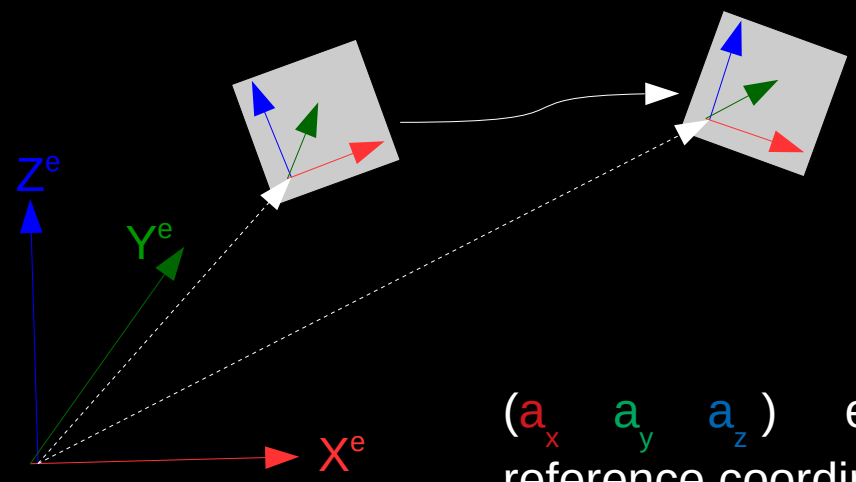
Trivial? It will help us
with rotations.
they will change!

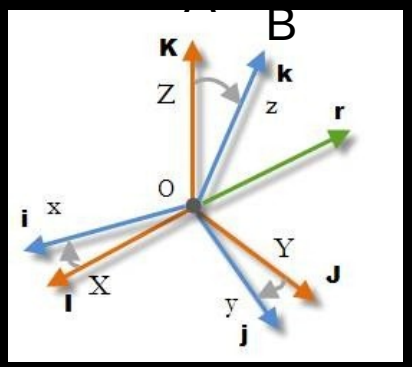
Unambiguous with 2 (or 3) Vectors



$(a_x \quad a_y \quad a_z)$ -> $(b_x \quad b_y \quad b_z)$

each a and each b is a vector



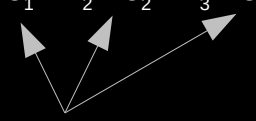


$(a_x \quad a_y \quad a_z)$ each one is a **vector** related to a reference coordinate system (e)

a_x , is the red vector .... etc.

same for b.

$V = v_1 * e_1 + v_2 * e_2 + v_3 * e_3$



R = ((1,0,0) (0,1,0) (0,0,1))

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

attitude of reference coordinate system (R)

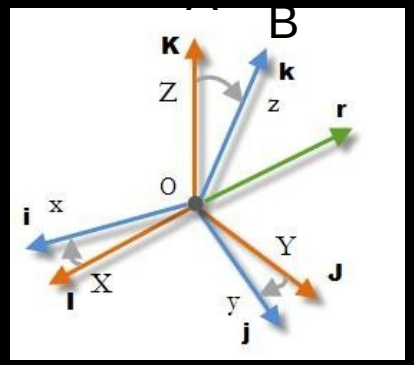
Julius-Maximillians-
UNIVERSITÄT
WÜRZBURG

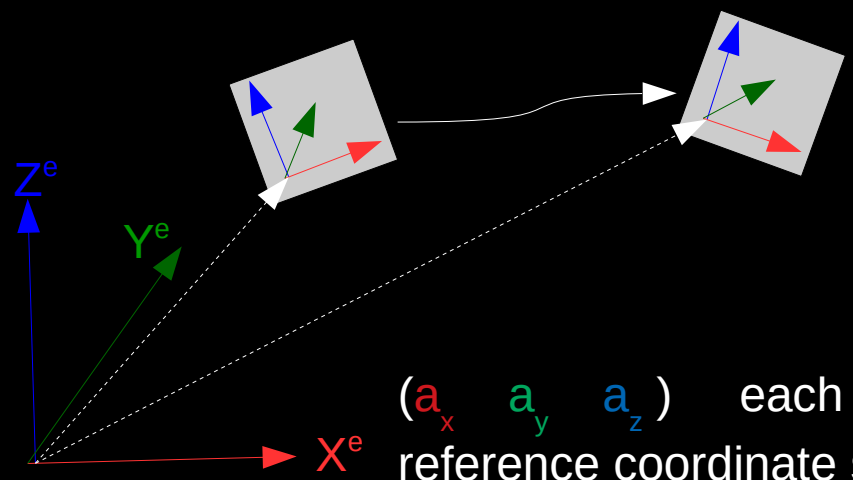
Aerospace Information Technology

Unambiguous with 2 (or 3) Vectors

$(a_x \quad a_y \quad a_z)$ -> $(b_x \quad b_y \quad b_z)$

each a and each b is a vector





$(a_x \quad a_y \quad a_z)$ each one is a **vector** related to a reference coordinate system (e)

a_x , is the red vector .... etc.

same for b.

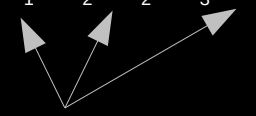
attitude of A

$$A = \begin{pmatrix} a_{x1} & a_{y1} & a_{z1} \\ a_{x2} & a_{y2} & a_{z2} \\ a_{x3} & a_{y3} & a_{z3} \end{pmatrix}$$

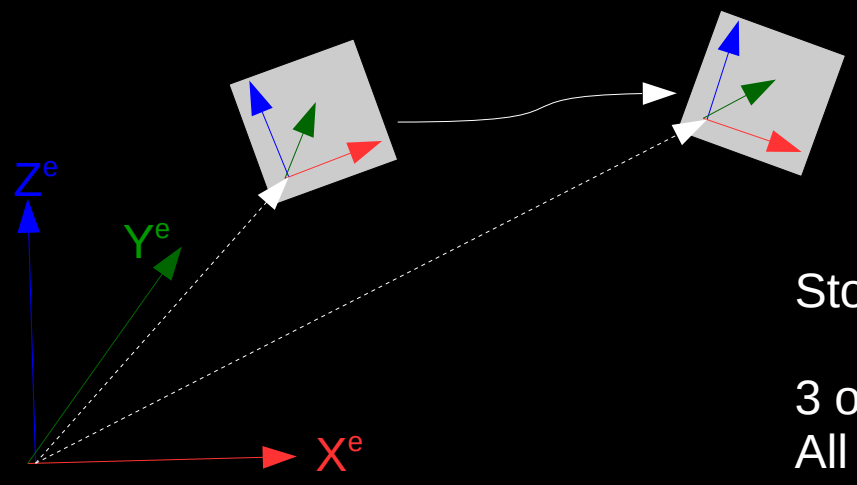
attitude of B

$$B = \begin{pmatrix} b_{x1} & b_{y1} & b_{z1} \\ b_{x2} & b_{y2} & b_{z2} \\ b_{x3} & b_{y3} & b_{z3} \end{pmatrix}$$

$V = v_1 * e_1 + v_2 * e_2 + v_3 * e_3$



Unambiguous with 2 (or 3) Vectors



Stop! We were talking about 2 not about 3!

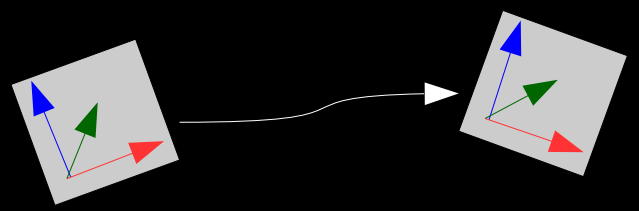
3 or 2 vectors?
All 3 are orthogonal
so 2 are enough, the
3rd one is Z = X*Y

And if they are not orthogonal?
At least 2 non-collinear, so
we may calculate all 3 orthogonal ones

DCM = Direction Cosinus Matrix

DCM

$(a_x \quad a_y \quad a_z) \quad \rightarrow \quad (b_x \quad b_y \quad b_z)$



-> direction-cosinus matrix :
a rotation matrix of scalar products from a to b

$$\cos(a_x \rightarrow b_x) = a_x \bullet b_x$$

$$\begin{pmatrix} a_x \bullet b_x & a_x \bullet b_y & a_x \bullet b_z \\ a_y \bullet b_x & a_y \bullet b_y & a_y \bullet b_z \\ a_z \bullet b_x & a_z \bullet b_y & a_z \bullet b_z \end{pmatrix}$$

Beware!
Each element, e.g. a_x is a 3D vector
and $a_x \bullet b_x$ is an scalar

attitude A

$(a_x \quad a_y \quad a_z)$

$$\begin{pmatrix} a_{x1} & a_{y1} & a_{z1} \\ a_{x2} & a_{y2} & a_{z2} \\ a_{x3} & a_{y3} & a_{z3} \end{pmatrix}$$

Or easier to read A → A'

$$\begin{pmatrix} X \bullet X' & X \bullet Y' & X \bullet Z' \\ Y \bullet X' & Y \bullet Y' & Y \bullet Z' \\ Z \bullet X' & Z \bullet Y' & Z \bullet Z' \end{pmatrix}$$

Beware!

Each element, e.g. a_x is a 3D vector

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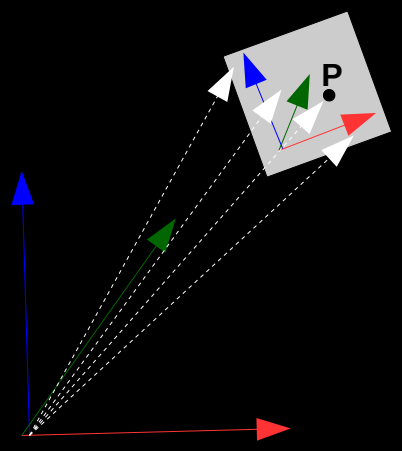


attitude'
3x3 Matrix

$$[X'Y'Z'] \times \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

attitude
3x3 Matrix

Or easier to read A → A'

$$\begin{pmatrix} X \bullet X' & X \bullet Y' & X \bullet Z' \\ Y \bullet X' & Y \bullet Y' & Y \bullet Z' \\ Z \bullet X' & Z \bullet Y' & Z \bullet Z' \end{pmatrix}$$

Why?

Rotation matrix A → A’

$$\begin{pmatrix} X \bullet X' & X \bullet Y' & X \bullet Z' \\ Y \bullet X' & Y \bullet Y' & Y \bullet Z' \\ Z \bullet X' & Z \bullet Y' & Z \bullet Z' \end{pmatrix}$$

attitude of A

$$(a_x \quad a_y \quad a_z)$$

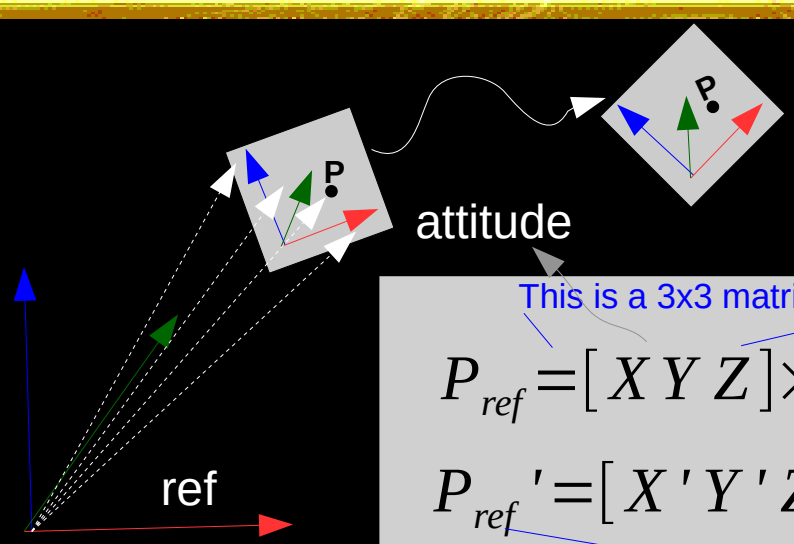
$$\begin{pmatrix} a_{x1} & a_{y1} & a_{z1} \\ a_{x2} & a_{y2} & a_{z2} \\ a_{x3} & a_{y3} & a_{z3} \end{pmatrix}$$

Let’s show it with a demo rotation

Beware!

Each element, e.g. X or X' is a 3D vector

$$\begin{pmatrix} X \bullet X' & X \bullet Y' & X \bullet Z' \\ Y \bullet X' & Y \bullet Y' & Y \bullet Z' \\ Z \bullet X' & Z \bullet Y' & Z \bullet Z' \end{pmatrix} = \underbrace{[X'Y'Z']}_{\text{attitude' 3x3 Matrix}} \times \underbrace{\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}}_{\text{attitude 3x3 Matrix}}$$



unit vectors relative to e

$$[X_e Y_e Z_e] \stackrel{\text{def}}{=} [X\,Y\,Z]$$

This is a 3x3 matrix!

$$P_{ref} = [X\,Y\,Z] \times P$$

$$P_{ref}' = [X'Y'Z'] \times P$$

back:

$$P = [X\,Y\,Z]^{-1} \times P_{ref}$$

$$P_{ref}' = [X'Y'Z'] \times [X\,Y\,Z]^{-1} \times P_{ref}$$

$$P_e' = [X'Y'Z'] \times \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \times P_e$$

Because (with orthogonal vectors) $A^{-1} = A^T$

$$A_{3,3} = \begin{pmatrix} 5 & 7 & 3 \\ 8 & 16 & 9 \\ 22 & 1 & 7 \end{pmatrix} \quad A^T_{3,3} = \begin{pmatrix} 5 & 8 & 22 \\ 7 & 16 & 1 \\ 3 & 9 & 7 \end{pmatrix}$$

3 x 3 Matrix 3 x 3 Matrix

Rotation Matrix

Direction Cosinus Matrix "**D**"

$$\begin{pmatrix} X \bullet X' & X \bullet Y' & X \bullet Z' \\ Y \bullet X' & Y \bullet Y' & Y \bullet Z' \\ Z \bullet X' & Z \bullet Y' & Z \bullet Z' \end{pmatrix}$$

e.g. I want to rotate vector W:

$$W' = \mathbf{D} * W$$

Or the position of W relative
to reference coordinate system
Then I say, D is the **attitude**

$$W_{reference} = D * W_{bodyframe}$$

Rotation Calculations

https://en.wikipedia.org/wiki/Rotation_formalisms_in_three_dimensions

- Rotation matrix — Around X,Y,Z, e.g. Dir. Cos. Matrix (DCM)
- Euler-Angle (incl. RPY *) — aviation, but ony for output
- Quaternions — around axis, space & comp. games

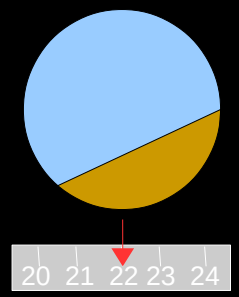
rotation vector, rotation angle

(* RPY = Roll, Pitch, Yaw)

Rotation Calculations

https://en.wikipedia.org/wiki/Rotation_formalisms_in_three_dimensions

2D representation



Rotation matrix $\begin{pmatrix} +0{,}999981452 & -0{,}005585025 & -0{,}002429486 \\ +0{,}005585008 & +0{,}999984404 & -0{,}000006789 \\ +0{,}002429496 & -0{,}000006981 & +0{,}999997049 \end{pmatrix}$

Better for sequences of rotations, micro integration and extreme rotations

Euler-Angle (incl. RPY *)

- Gimball lock
- ambiguous (at nick 90°)
- etc. etc.

But... the pilot does not want rotation axis, but Roll, Pitch, Yaw

Quaternions (0.2834, (0.234, 0.7543, 0.5822))

Better for sequences of rotations, micro integration and extreme rotations

Suggestion: We calculate everything as quaternion and for the pilot we convert into RPY.